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LOOKING BACK AT 2020, WE CAN BE CERTAIN THAT IT'S GOING TO BE A MEMORABLE ONE. The global pandemic certainly made things worse for a lot of us and even for the ICT industry. However, we saw a quick recovery or at least an attempt to do so. Many hardware companies had promises to keep for their investors and we saw several paper launches. The sad reality being that the hardware was actually pretty good, the only shortcoming is that there isn't enough silicon for everybody.

The big three PC hardware companies – Intel, AMD and NVIDIA – all made decent launches but stocks were hard to find anywhere. There's also the fact that India isn't a big market for any of these companies in the DIY desktop hardware space and thus, we don't even see proper allocation of stocks for our country. And the companies which rely on the aforementioned three are facing similar problems.

Microsoft and Sony don't have enough consoles to go around. The Xbox Series X was announced early on in the country and they even had a short pre-order run for the console but stocks ran out super quick and resupply is literally months away. Sony didn't even bother launching their consoles in India, they're about to have pre-orders from the 12th of Jan but stocks will again be severely limited. The problem here is compounded due to lack of silicon and India not being a priority market.

Production issues aside, the big three have larger concerns looming on the horizon as more companies have started exploring custom silicon for their own needs. Actually, only Intel and AMD need worry since NVIDIA now owns ARM and most of the custom silicon being explored is using ARM cores. Apple already made the move to ARM for their laptops and they've bought out a good chunk of TSMC's 5nm and 3nm capacities for their custom silicon. This shows that they are confident about their custom silicon. Even Microsoft is working on in-house chips using ARM cores for their servers which are currently running mostly on Intel's chips. Amazon has been working on their own in-house chips for quite a while and they're another of Intel's biggest clients in the server space. Every company has its own reasons

for exploring in-house designs but chief among them is cost. Server processors are ridiculously expensive and the cloud business is only set to grow so it makes sense to invest in in-house designs that will help reduce some of the largest infrastructure expenses that any cloud company has to bear.

RISC-V has also seen greater adoption. And this concerns all x86 and ARM chip makers as RISC-V is open-source and requires no royalties to be paid to any IP holder. As a result, many research institutes have picked up RISC-V for their in-house projects. Here in India, IIT Madras is reportedly working on six new designs across varying performance levels. Some of which will power future Electronic Voting Machines (EVMs).

Quite a few folks have prompted the question – Was 2020 the best year for tech? In the recent past, you could say so. A large part of the digitisation happened due to the pandemic and the massive amount of IP launches would also indicate a decent growth. However, this was nothing compared to the early days of the x86 boom during the early '90s. Back then I was just starting off with PCs on my first ever Pentium II machine but that's when the actual boom happened. Tech that launched was already obsolete over the course of two quarters. Monumental performance growths were being achieved thanks to disruptive technologies that were coming out from all corners. Back then, companies were just starting to put out incremental product launches just to stretch out an IP for a longer time and make more money. The 3-10% growth trend that we see gen-on-gen was yet to begin. Now, we've reached a point where processing power is available in plenty so that kind of disruptive growth no longer makes sense. Does that make the ecosystem jaded? One could argue that it does, but there's always excitement around new launches and new entrants in the space. We don't know where the next disruptive tech will come from but we're all very excited for one to appear. This month's cover story explores the different domains from where a disruptive tech is likely to emerge from. We've covered 20 of the biggest trends to watch for in 2021. We'll certainly be watching with bated breath, I hope you're equally excited for the same. 



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